

**KWAME NKRUMAH UNIVERSITY OF SCIENCE AND
TECHNOLOGY, KUMASI**



**HOSPITAL CHATBOT FOR COLOR-CODED CLINICAL
PATHWAYS**

BY
QUIAGRAINE SAMUEL
&
TWUM SAMUEL

A THESIS SUBMITTED TO THE DEPARTMENT OF COMPUTER
SCIENCE, KWAME NKRUMAH UNIVERSITY OF SCIENCE AND
TECHNOLOGY IN PARTIAL FULFILLMENT OF THE REQUIREMENT
FOR THE DEGREE OF BACHELOR OF SCIENCE

JULY 2026

DECLARATION

We hereby declare that this submission is our own work towards the B.Sc. degree and that, to the best of our knowledge, it contains no material previously published by another person nor material which has been accepted for the award of any other degree of the University, except where due acknowledgement has been made in the text.

Quaigraine Samuel

Student

Signature / Date

Twum Samuel

Student

Signature / Date

Certified by:

Supervisor

Signature / Date

DEDICATION

This work is dedicated to patients and frontline triage staff in Ghanaian emergency departments, whose safety depends on timely, colour-coded clinical pathways.

ABSTRACT

Emergency departments in low- and middle-income countries (LMICs) face overcrowding, delayed first assessment, and a failure we call *acuity discontinuity*: the South African Triage Scale (SATS) colour assigned at the desk often stops guiding care after the patient leaves triage. This thesis presents a hospital chatbot that maps an unstructured chief complaint to a SATS acuity level and colour, then attaches a concrete clinical pathway—destination, target response time, and actions.

Part 1 implements a Digital Acuity Engine: a medical gate filters non-clinical text; optional OpenMed named-entity enrichment surfaces diseases and drugs; a fine-tuned BioBERT classifier predicts acuity levels 1–5; a fixed map f_{SATS} yields Red, Orange, Yellow, or Green. On an 8,000-sample stratified holdout, baseline BioBERT achieved 99.92% colour-routing accuracy and 98.14% Level-1 (Red) recall.

Part 2 defines protocol-mapped pathways $P(C)$ for each colour (including Green diversion from the acute ED and a Blue dignity protocol), and implements a multi-signal fusion controller f_{fusion} over NLP colour, optional TEWS vitals, rule-based discriminators, and a *tabular* Bayesian fallback (not a full Bayesian network). Pathways attach to the **fused** colour via `/fuse`; text-only `/predict` remains NLP-only for compatibility.

The contribution is a defensible LMIC chatbot design and prototype (Curatio) that shows NLP can drive colour-coded pathways while optional vitals and safety layers prevent under-triage when language and physiology disagree.

Keywords: SATS; clinical pathways; BioBERT; triage chatbot; LMIC; acuity discontinuity.

ACKNOWLEDGMENT

We thank our supervisor and the Department of Computer Science, KNUST, for guidance. We acknowledge the South African Triage Group for the SATS framework that underpins colour coding in this work, and the open research community behind BioBERT and related clinical NLP tools used in the Curatio prototype.

CONTENTS

DECLARATION	i
DECLARATION	i
DEDICATION	ii
DEDICATION	ii
ACKNOWLEDGMENT	iv
ACKNOWLEDGMENT	iv
ABBREVIATION	vii
ABBREVIATIONS	viii
LIST OF TABLES	ix
LIST OF FIGURES	x
1 INTRODUCTION	1
1.1 Background of the study	1
1.2 Problem statement	2
1.3 Aim and objectives	2
1.3.1 General objective	2
1.3.2 Specific objectives	2
1.4 Research questions	3
1.5 Scope and limitations	4

1.6	Contributions	4
1.7	Organisation of the thesis	5
2	LITERATURE REVIEW	6
2.1	Introduction	6
2.2	Emergency overcrowding and the Green Tag Burden	6
2.3	SATS in LMIC and Ghanaian practice	7
2.4	Triage Early Warning Score (TEWS) — parallel vitals layer . . .	7
2.5	Tabular Bayesian fallback (not a full BN)	7
2.6	Clinical NLP and BioBERT	8
2.7	Conversational agents and digital triage tools	8
2.8	From colour labels to clinical pathways	8
2.9	Summary of the gap	9
3	METHODOLOGY	10
3.1	Overview	10
3.2	Data and fine-tuning	10
3.3	Medical gate	10
3.4	OpenMed enrichment	11
3.5	BioBERT inference	11
3.6	Colour mapping	11
3.7	Pathway protocols $P(C)$	12
3.8	TEWS, discriminators, and tabular Bayes	12
3.9	Evaluation protocol	12
3.10	System prototype (Curatio)	13
3.11	What is not in the methodology	13
4	RESULTS AND DISCUSSION	14
4.1	Holdout classification performance	14
4.2	Errors and Red safety	14
4.3	Calibration and confidence	15

4.4	Medical gate and OpenMed (qualitative / system tests)	15
4.5	Pathway and fusion scenarios (Part 2)	15
4.6	Limitations	16
4.7	Discussion relative to hybrid TEWS–Bayes designs	16
5	CONCLUSION AND FUTURE WORK	17
5.1	Conclusion	17
5.2	Answers to research questions (summary)	17
5.3	Contributions restated	18
5.4	Future work	18
	REFERENCES	19
	APPENDIX	20
5.5	Formal maps	20
5.6	Repository pointers	20
5.7	Evaluation artefacts	20

ABBREVIATIONS

SATS South African Triage Scale

TEWS Triage Early Warning Score

ED Emergency Department

OPD Outpatient Department

LMIC Low- and Middle-Income Country

KATH Komfo Anokye Teaching Hospital

NLP Natural Language Processing

NER Named Entity Recognition

BioBERT Biomedical Bidirectional Encoder Representations from Transformers

SMOTE Synthetic Minority Over-sampling Technique

PRISMA Preferred Reporting Items for Systematic Reviews and Meta-Analyses

API Application Programming Interface

LIST OF TABLES

3.1	Protocol-mapped colour pathways (chatbot Part 2).	12
4.1	Baseline vs SMOTE BioBERT on stratified holdout.	14

LIST OF FIGURES

CHAPTER 1

INTRODUCTION

1.1 Background of the study

In high-functioning emergency care, **triage** prioritises patients by urgency so that the right patient receives the right care at the right time. The **South African Triage Scale (SATS)**—widely used in LMIC settings including Ghana—assigns colour codes: Red (immediate), Orange (very urgent, typically within 10 minutes), Yellow (urgent, typically within 60 minutes), Green (non-urgent), and Blue (deceased / dignity pathway).

These colours are meant to trigger **clinical pathways**: standardised roadmaps of destination, diagnostics, and time targets. In practice, however, colour often becomes a label that ends at the triage desk. Once a patient moves to imaging, a ward, or a generic queue, the hospital ecosystem may “forget” their acuity. We call this failure **acuity discontinuity**.

Compounding the problem is the **Green Tag Burden**: non-urgent patients occupy acute emergency capacity. Pre-triage queues may also delay first assessment for hours. Patients arriving via digital channels can describe symptoms in free text long before six TEWS vitals are available.

This project develops a **hospital chatbot for color-coded clinical pathways**. The chatbot collects a chief complaint, predicts SATS acuity and colour with clinical NLP, optionally fuses TEWS vitals with discriminators and a tabular Bayesian fallback, and attaches an operational pathway card to the fused colour.

1.2 Problem statement

Resource-constrained hospitals face:

1. **Pre-triage bottleneck** — delay before any colour is assigned.
2. **Cognitive load** — manual TEWS calculation under pressure risks under- and over-triage.
3. **Static triage** — one-shot colour without digital timers or re-assessment prompts.
4. **Lost urgency** — Orange/Yellow protections are not tracked after the ED desk.
5. **Text-only intake** — vitals are often missing at home or in the queue, so a language-based acuity signal is needed at the door.

Generic consumer symptom checkers are not aligned to SATS hospital pathways or local operational routing (e.g. Polyclinic vs acute ED).

1.3 Aim and objectives

1.3.1 General objective

To design and prototype a chatbot-mediated Digital Acuity Engine that maps unstructured chief complaints to SATS colour-coded clinical pathways for LMIC hospital intake.

1.3.2 Specific objectives

1. Fine-tune and evaluate a BioBERT classifier that maps chief complaints to acuity levels 1–5.

2. Map acuity to SATS colour via a fixed function f_{SATS} and report colour-routing performance.
3. Define protocol-mapped pathways $P(C)$ for Red, Orange, Yellow, Green, and Blue.
4. Implement a chatbot/API prototype (Curatio) with medical gating and optional clinical NER enrichment.
5. Document and implement a fusion architecture f_{fusion} with pathways $P(C)$ on fused colour (TEWS + discriminators + tabular Bayes), keeping full BN learning out of scope.

1.4 Research questions

- RQ1** Can unstructured chief complaints be mapped to SATS acuity levels 1–5 with clinically usable accuracy?
- RQ2** Does correct acuity prediction imply correct colour assignment under fixed f_{SATS} ?
- RQ3** How should colours translate into pathways (destination, T_{max} , actions)?
- RQ4** Can a chatbot deliver pre-triage acuity and pathway guidance before the nurse desk?
- RQ5** How well does the system reject non-clinical input?
- RQ6** Does OpenMed NER enrichment improve interpretability without replacing the acuity model?
- RQ7** How does class imbalance affect Level-1 (Red) recall, and what mitigation is appropriate?
- RQ8** How overconfident is softmax output, and what does that imply for nurse oversight?

RQ9 Can Green-pathway diversion be specified to reduce acute ED congestion?

RQ10 Does f_{fusion} assign a safer SATS colour than C_{NLP} alone when TEWS and language disagree?

RQ11 Are pathway cards $P(C)$ correctly attached to fused colours in end-to-end API tests?

1.5 Scope and limitations

In scope: $\text{text} \rightarrow \text{BioBERT} \rightarrow C_{\text{NLP}}$; optional TEWS, discriminators, tabular Bayes; max-urgency fusion; pathway protocols $P(C)$; medical gate; OpenMed enrichment; holdout + fusion scenario evaluation; Curatio prototype (`/predict`, `/fuse`).

Out of scope: full Bayesian networks (pgmpy); learned discriminator neural heads; prospective KATH clinical trial; full HIS integration; EMR-calibrated Bayes priors.

TEWS algebra, rule discriminators, and tabular $P(C_k | E)$ are **implemented** in Part 2 (`triage-fusion/`); full BN structure learning remains future work.

1.6 Contributions

- An implemented NLP Digital Acuity Engine with measured holdout performance.
- A protocol table linking SATS colours to hospital pathways (including Green diversion).
- A fusion controller f_{fusion} (max-urgency) with TEWS, discriminators, and tabular Bayes, exposed as `POST /fuse`.
- Open framing documents and a first-draft thesis structure for continued writing.

1.7 Organisation of the thesis

Chapter 2 reviews SATS, overcrowding, and triage NLP. Chapter 3 presents methodology. Chapter 4 reports results and discusses pathways. Chapter 5 concludes and states future work.

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction

This chapter situates the project in emergency triage practice (SATS), LMIC overcrowding, clinical NLP for triage, and conversational systems. Systematic-review reporting can follow PRISMA 2020 templates archived under `systematic-review-prisma/`. Detailed paper notes belong in `research-findings/`.

2.2 Emergency overcrowding and the Green Tag Burden

Emergency departments worldwide face overcrowding. A structural driver is the influx of non-urgent patients into acute facilities—the so-called Green Tag Burden. In LMICs, understaffing and limited diagnostics worsen the competition for beds and clinician time. Digital diversion of Green patients to OPD, Minors, or Polyclinic streams is therefore not only a convenience feature but a flow-management requirement for colour-coded pathways.

Waiting time interacts with colour assignment and outcomes; literature on waiting time and triage colour codes informs why pathway timers ($T_{\max}$) matter after a colour is assigned.

2.3 SATS in LMIC and Ghanaian practice

SATS combines physiological early warning (TEWS) with clinical discriminators that can override mild vital-sign scores when high-risk symptoms are present (e.g. central chest pain). Implementation studies at Komfo Anokye Teaching Hospital (KATH) and in other LMIC ambulance/ED settings demonstrate that SATS is operationally relevant beyond South Africa.

Implication for this thesis: Colour codes already exist in hospital policy. The research gap is not inventing colours, but (i) assigning them from free-text complaints before full vitals, and (ii) binding colours to pathway protocols so acuity does not disappear after the desk.

2.4 Triage Early Warning Score (TEWS) — parallel vitals layer

TEWS aggregates mobility, respiratory rate, heart rate, temperature, AVPU, and trauma into a sum that maps to colour bands. It is central to bedside SATS. Chatbot intake may lack complete vitals, so TEWS is an **optional parallel layer**: when vitals are supplied, C_{TEWS} votes in f_{fusion} ; when missing, partial TEWS is flagged and cannot silently downgrade urgency.

2.5 Tabular Bayesian fallback (not a full BN)

Multimodal Bayesian triage models appear in the literature as ways to combine uncertain evidence. This thesis implements a **tabular** fallback $P(C_k \mid E)$ with scenario-keyed priors and likelihoods (e.g. chest pain + partial vitals), not structure learning or pgmpy. Full Bayesian networks remain related work / future calibration research. Softmax C_{NLP} remains the primary language signal; tabular Bayes is invoked when TEWS is incomplete, NLP confidence is low, or layers

conflict.

2.6 Clinical NLP and BioBERT

Transformer models—BERT and domain-adapted BioBERT—enable contextual embeddings for biomedical text. Deep attention models have been applied to ED triage notes. Reviews of NLP at ED triage (e.g. Stewart and colleagues) summarise opportunities and risks. Classification with a fixed label set reduces open-ended generative hallucination risk compared with unconstrained large language models.

Our pipeline fine-tunes BioBERT for five-class acuity prediction on chief complaints, optionally enriched with OpenMed NER tags for diseases and medications.

2.7 Conversational agents and digital triage tools

Symptom-assessment applications and ED prediction chatbots show that conversational NLP can support triage-like advice. Validation studies emphasise safety, under-triage monitoring, and comparison with conventional triage. Our chatbot is constrained: it outputs acuity, colour, and pathway cards aligned to SATS—not free-form diagnosis essays.

2.8 From colour labels to clinical pathways

Clinical pathways are step-by-step care roadmaps. Synopsis and clinical protocol materials in this project define Five-Path logic: Red resuscitation, Orange countdown, Yellow parallel standing orders, Green diversion, Blue dignity. The literature supports treating colour as operational, not decorative.

2.9 Summary of the gap

Prior art provides SATS colours, TEWS bedside scoring, and growing NLP triage evidence. Missing for LMIC chatbot intake is an integrated story: accurate text→acuity→colour **plus** optional vitals/discriminator/Bayes fusion **plus** explicit pathway protocols on the fused colour. Chapters 3–4 address that gap.

CHAPTER 3

METHODOLOGY

3.1 Overview

The methodology has three layers:

1. **Digital Acuity Engine (Part 1):** $\text{text} \rightarrow \text{acuity } \hat{c} \rightarrow C_{\text{NLP}} = f_{\text{SATS}}(\hat{c})$.
2. **Fusion controller:** optional TEWS, discriminators **D**, and tabular Bayes combine with C_{NLP} via max-urgency f_{fusion} .
3. **Pathway protocols:** fused $C \mapsto P(C) = (T_{\text{max}}, \text{dest}, \text{actions}, \text{escalation})$.

Text-only `/predict` remains NLP-only; multi-signal intake uses `/fuse`.

3.2 Data and fine-tuning

Chief-complaint text paired with acuity labels was used to fine-tune BioBERT for sequence classification over five classes. Training artefacts and notebooks live under `fine-tuned-biobert/`. A stratified 10% holdout ($n = 8,000$) was reserved for evaluation. An optional SMOTE-augmented training variant was compared for minority-class (Red) recall.

3.3 Medical gate

Before acuity prediction, a clinical relevance gate scores whether the input is a medical chief complaint. Non-clinical text (greetings, shopping, sports, etc.) is rejected with a dynamic message and category (e.g. `non_clinical_topic`).

Default threshold: 0.35. This protects the triage model from nonsense inputs and answers RQ5.

3.4 OpenMed enrichment

When enabled, OpenMed NER extracts disease/condition and medication spans, with simple negation heuristics. Optional entity prefixes of the form [DISEASE: ...] / [DRUG: ...] may be prepended before BioBERT tokenisation. OpenMed does **not** predict acuity; BioBERT remains the classifier (RQ6).

3.5 BioBERT inference

Let X be the (possibly enriched) complaint. BioBERT produces logits $\mathbf{z} \in \mathbb{R}^5$. Softmax yields

$$\hat{y}_c = \frac{e^{z_c}}{\sum_{j=1}^5 e^{z_j}}, \quad \hat{c} = \arg \max_c \hat{y}_c, \quad \text{confidence} = \max_c \hat{y}_c.$$

Training minimises cross-entropy against nurse acuity labels.

3.6 Colour mapping

$$C = f_{\text{SATS}}(\hat{c}) = \begin{cases} \text{Red} & \hat{c} = 1 \\ \text{Orange} & \hat{c} = 2 \\ \text{Yellow} & \hat{c} = 3 \\ \text{Green} & \hat{c} \in \{4, 5\} \end{cases}$$

Implemented in the ML service as a constant lookup. Under this fixed map, colour-routing accuracy equals acuity accuracy (RQ2).

3.7 Pathway protocols $P(C)$

Table 3.1: Protocol-mapped colour pathways (chatbot Part 2).

Colour	$T_{\max}$	Destination	Actions (summary)
Red	0 min	Resuscitation	Code Red; bypass registration
Orange	10 min	Acute bed	Countdown; escalate if unclaimed
Yellow	60 min	Urgent stream	ED Standing orders in parallel
Green	Routine	OPD / Minors / Polyclinic	Divert from acute ED
Blue	Dignity	Mortuary pathway	Silent social-work protocol

Blue is a dignity protocol outside the five-class acuity head. Pathways attach to **fused** colour from `/fuse` (or to C_{NLP} for text-only `/predict` demos).

3.8 TEWS, discriminators, and tabular Bayes

When vitals are present, TEWS scores f_1 – f_6 yield T and C_{TEWS} . Rule/OpenMed discriminators yield C_{disc} . Tabular Bayes yields C_{Bayes} when incomplete TEWS, low NLP confidence, layer conflict, or `force_bayes` applies. Fusion:

$$\text{ord}(C) \geq \max\{\text{ord}(C_{\text{disc}}), \text{ord}(C_{\text{TEWS}}), \text{ord}(C_{\text{Bayes}}), \text{ord}(C_{\text{NLP}})\}.$$

Incomplete TEWS Green is suppressed so missing vitals cannot downgrade a higher NLP/discriminator signal.

3.9 Evaluation protocol

On the holdout set we report accuracy, precision, recall, F1, macro-F1, per-class support, confusion matrices, mean confidence, and calibration plots. Level-1

recall is emphasised as a safety metric (RQ7). Softmax overconfidence informs nurse-in-the-loop guidance (RQ8). Fusion scenarios (chest pain, critical vitals, Green diversion) are tabulated for RQ10–RQ11.

3.10 System prototype (Curatio)

- **ML service:** FastAPI — `/predict`, `/fuse`, `/analyze`, `/deidentify`, voice intake.
- **API gateway:** Express proxies to the ML service.
- **Client:** React triage test page (vitals + layer audit) and presentation dashboard.

Default predict path enables the medical gate and OpenMed enrichment; both can be disabled for ablation demos. OpenMed NER is timeout-bounded so model downloads cannot block acuity forever.

3.11 What is not in the methodology

Full Bayesian network structure learning (pgmpy), EMR-calibrated priors, and a learned multi-label discriminator head are excluded. Prospective clinical trials remain future work.

CHAPTER 4

RESULTS AND DISCUSSION

4.1 Holdout classification performance

Evaluation used a stratified 10% holdout of $n = 8,000$ complaints. Summary metrics:

Table 4.1: Baseline vs SMOTE BioBERT on stratified holdout.

Metric	Baseline	SMOTE
Accuracy	0.9992	0.9985
Macro-F1	0.9977	0.9954
Recall L1 (Red)	0.9814	1.0000
Mean confidence	0.9997	0.9992
% confidence = 1.0	79.05%	82.99%
Latency p50 (ms/sample)	100.3	95.5

RQ1: Baseline BioBERT maps complaints to acuity with very high holdout accuracy. **RQ2:** With fixed f_{SATS} , colour-routing accuracy matches acuity accuracy ($7,994/8,000 = 0.9992$).

Default inference uses the baseline checkpoint (slightly higher overall accuracy/macro-F1). SMOTE maximises Red recall on this holdout at a small cost to overall metrics—a clinical trade-off for RQ7.

4.2 Errors and Red safety

All six baseline errors on the reported holdout involved true Level 1 (Red) cases routed to a less urgent colour ($TP = 316$, $FN = 6$, $FP = 0$ for L1). This underscores why Red recall and nurse oversight matter even when overall accuracy is high.

4.3 Calibration and confidence

Mean confidence is extremely high; about 79% of baseline predictions have confidence exactly 1.0. Calibration plots show overconfidence. **RQ8:** pathway assignment should remain advisory to clinicians; low-confidence or safety-critical presentations warrant bedside TEWS and human confirmation. Overconfidence is **not** solved by adding Bayesian fusion in this thesis; it motivates process controls.

4.4 Medical gate and OpenMed (qualitative / system tests)

Non-clinical prompts (e.g. sports, streaming) are rejected with explanatory messages when the gate is enabled (RQ5). OpenMed enrichment surfaces disease and drug spans (e.g. chest pain, dyspnoea, metformin) alongside acuity, improving interpretability without replacing BioBERT (RQ6).

4.5 Pathway and fusion scenarios (Part 2)

POST /fuse returns layer votes and a pathway on fused colour. Representative cases (unit-tested; NLP colour mocked where noted):

- **Chest pain + partial vitals:** C_{NLP} Orange, C_{TEWS} Yellow ($T = 4$), C_{disc} Orange, C_{Bayes} Orange $\approx 0.89 \rightarrow$ fused **Orange**, $T_{\text{max}} = 10$ min.
- **Mild language + critical vitals:** fusion follows TEWS urgency (Red/Orange) over mild NLP wording (RQ10).
- **Green diversion:** mild rash + normal full vitals $\rightarrow$ Green $\rightarrow$ OPD/Polyclinic (RQ9).

Full table: triage-fusion/fusion-scenario-results.md. Text-only /predict remains NLP-only (Scenario 8).

4.6 Limitations

- Holdout is in-distribution relative to training complaints; live KATH language may differ.
- Softmax overconfidence limits trustworthy autonomous routing.
- Pathway timers and HIS escalation are designed, not fully instrumented as hospital middleware.
- Tabular Bayes priors are clinician-elicited / literature-plausible, not EMR-calibrated.
- No prospective clinical trial is claimed in this draft.

4.7 Discussion relative to hybrid TEWS–Bayes designs

Hybrid designs that fuse TEWS, Bayes, and NLP are now implemented in Curatio as a **max-urgency** controller with a *tabular* Bayes fallback. This is deliberately simpler than a full BN: optional vitals at intake escalate safely when present, while text-only patients still receive C_{NLP} pathways. Richer BN calibration remains future work if prospectively justified.

CHAPTER 5

CONCLUSION AND FUTURE WORK

5.1 Conclusion

This thesis addressed **Hospital Chatbot for Color-Coded Clinical Pathways** in two parts. Part 1 delivered a working Digital Acuity Engine: medical gating, optional clinical NER, fine-tuned BioBERT acuity prediction, and deterministic SATS colour mapping, with strong holdout colour-routing performance. Part 2 specified protocol-mapped pathways and implemented multi-signal fusion f_{fusion} (TEWS, discriminators, tabular Bayes, max-urgency) so each fused colour carries destination, time target, and actions—including Green diversion and Blue dignity—thereby attacking acuity discontinuity.

Softmax still yields the primary language distribution C_{NLP} ; pathways on /fuse attach to the **fused** colour. That answers RQ10–RQ11 with a practical LMIC intake stack that can escalate when vitals or discriminators demand it, without claiming a full EMR-trained Bayesian network.

5.2 Answers to research questions (summary)

- **RQ1–RQ2:** Yes—high holdout acuity/colour accuracy under fixed f_{SATS} .
- **RQ3, RQ9:** Pathways defined as protocol table including Green diversion.
- **RQ4:** Curatio chatbot/API demonstrates pre-desk delivery of acuity, fusion audit, and pathway cards.
- **RQ5–RQ6:** Gate and OpenMed behave as intended in system tests (NER timeout-hardened).

- **RQ7–RQ8:** Red recall and calibration overconfidence require safety-aware deployment.
- **RQ10:** Fusion max-urgency is at least as urgent as NLP when TEWS/discriminators/Bayes escalate.
- **RQ11:** `/fuse` attaches $P(C)$ to fused colour in API/unit tests.

5.3 Contributions restated

1. Implemented and evaluated text-to-SATS-colour triage NLP.
2. Formalised colour-coded clinical pathway protocols for chatbot output.
3. Implemented TEWS + discriminators + tabular Bayes + max-urgency fusion (`/fuse`).

5.4 Future work

- Instrument pathway timers and HIS escalation in live hospital middleware.
- Calibrate tabular Bayes priors on local EMR / prospective KATH data.
- Prospective evaluation with clinicians; measure under-triage and time-to-provider.
- Domain adaptation to Ghanaian informal English / Twi-translated complaints.
- Optional: full BN structure learning or a learned discriminator head—only if justified against the tabular baseline.

REFERENCES

APPENDIX

5.5 Formal maps

Acuity to colour:

$$C = f_{\text{SATS}}(\hat{c}).$$

Pathway:

$$P(C) = (T_{\max}(C), \text{dest}(C), \text{actions}(C), \text{escalation}(C)).$$

5.6 Repository pointers

- Prototype: `curatio/`
- Model training: `fine-tuned-biobert/`
- Metrics: `results/eval_outputs/`
- Framing: `project-framing/`
- Reading notes: `research-findings/`
- PRISMA templates: `systematic-review-prisma/`

5.7 Evaluation artefacts

Confusion matrices and calibration plots: `results/eval_outputs/baseline_*.png`,
`smote_*.png`.